Gluing (pasting) lemma:

Let $X = A \cup B$, where $A, B \subset X$ are closed. Suppose $f: A \to Y$, $g: B \to Y$ are continuous (in the subspace topology) and that $f(x) = g(x)$ for all $x \in A \cap B$. Define $h: X \to Y$ by $h(x) = \begin{cases} f(x), & x \in A \\ g(x), & x \in B \end{cases}$. Then h is continuous.

Pf: Let $C \subset Y$ be closed. Then $h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C)$. Now $f^{-1}(C) \subset A$, $g^{-1}(C) \subset B$ are closed in A and B, resp., so also in $X \Rightarrow h^{-1}(C) \subset X$ is closed.

(Recall: If $C \subset Y \subset X$ is closed then there is closed $C' \subset X$ s/t $C = C' \cap Y$. Thus if $Y \subset X$ is closed then $C \subset X$ is closed.)

Ex: A loop at $x_0 \in X$ is a cont. map $\nu: [0,1] \to X$ s/t $\nu(0) = \nu(1) = x_0$.

Now let ν_1, ν_2 be two such curves.

Define $\nu_1 * \nu_2(t) = \begin{cases} \nu_1(2t), & 0 \le t \le \frac{1}{2} \\ \nu_2(2t-1), & \frac{1}{2} \le t \le 1 \end{cases}$

Then $\nu_1 * \nu_2$ is cont.

§ 3

Let $A \subset X$ be a subset. A collection $\{B_\alpha\}_{\alpha \in \mathcal{A}}$ of subsets of X forms a cover for A if $A \subset \bigcup_{\alpha \in \mathcal{A}} B_\alpha$.

Ex: $B_n = (\frac{1}{n}, n) \subset \mathbb{R}$. $A = ?$

Say $C \subset X$ is compact if every open cover $\{O_\alpha\}_{\alpha \in \mathcal{A}}$ of C contains a finite sub cover, that is, there are $O_1, \dots, O_n \in \{O_\alpha\}_{\alpha \in \mathcal{A}}$ such that $C \subset O_1 \cup \dots \cup O_n$.

Ex: Give X the discrete topology. Find all compact subsets of X.

Ex: Let $C = (0, 1] \subset \mathbb{R}$. Is C compact?

Ex: $\mathbb{R}^n$ is not cmpt, $B_1(0) \setminus \{0\} \subset \mathbb{R}^n$ is not cmpt.

A space X is Hausdorff if given $p, q \in X$, $p \neq q$, then there are nbhds O_p of p and O_q of q s/t $O_p \cap O_q = \phi$.

Prop: Any metric space is Hausdorff.

Example: Non Hausdorff space ..._____•...

Prop: Let $Y \subset X$. Then Y is cmpt in the subspace topology $\iff$ Y is cmpt as a subset of X.

Pf: $\Rightarrow$ Let $O_\alpha \subset X$ be an open cover of Y. Then $Y \cap O_\alpha \subset Y$ is an open cover of Y in the subspace topology $\Rightarrow$ $Y \subset \bigcup_{i=1}^{k} Y \cap O_i$ $\Rightarrow$ $Y \subset \bigcup_{i=1}^{k} O_i$, so Y is cmpt in X.

$\Leftarrow$ Let $P_\alpha \subset Y$ be an open cover of Y. Then $P_\alpha = Y \cap O_\alpha$, $O_\alpha \subset X$ open. Now some $\{O_i\}_{i=1,\dots,k}$ covers Y so $\{P_i = O_i \cap Y\}_{i=1,\dots,k}$ covers Y $\Rightarrow$ Y is cmpt in X.

Prop: Let X be Hausdorff and $C \subset X$ cmpt. Then C is closed.

Pf: Let $p \in \neg C$. Then for every $q \in C$ there exist open nbhds $O_q \ni p$ and $P_q \ni q$ such that $O_p \cap P_q = \emptyset$

Now $C \subset \bigcup_{q \in C} P_q$ so by cmptness, there are $P_{q_1}, \ldots, P_{q_n}$ s/t $C \subset P_{q_1} \cup \ldots \cup P_{q_n}$. Then $p \in O_{q_1} \cap \ldots \cap O_{q_n}$ which is open and does not intersect C. So $\neg C$ is open $\Rightarrow$ C is closed

Prop: Let $C \subset X$ be cmpt and $f: X \to Y$ continuous. Then $f(C) \subset Y$ is cmpt.

Pf: Let $\{O_\alpha\}$ be an open cover of $f(C)$. Then $\{f^{-1}(O_\alpha)\}$ is an open cover of C so there is a finite $\{f^{-1}(O_1), \ldots, f^{-1}(O_n)\}$ covering C

Now $f(C) \subset f(f^{-1}(O_1) \cup \ldots \cup f^{-1}(O_n))$

$$= f(f^{-1}(O_1)) \cup \ldots \cup f(f^{-1}(O_n))$$

$$\subset O_1 \cup \ldots \cup O_n.$$

Prop. Let $\mathcal{C} \subset X$ be cmpt and let $K \subset \mathcal{C}$ be closed in X. Then K is cmpt.

Pf: Let $\{O_\alpha\}$ be an open cover of K. Then $\{O_\alpha\} \cup \{\neg K\}$ is an open cover of $\mathcal{C}$. $\mathcal{C}$ is cmpt so some finite $O_{\alpha_1}, \dots, O_{\alpha_n}, \neg K$ cover it. Then $O_{\alpha_1}, \dots, O_{\alpha_n}$ is a finite cover of K.

Prop. Let $K \subset X$ be cmpt in a metric space X. Then K is bounded, that is, there is $M < \infty$ s/t $d(p,q) < M$ for all $p, q \in K$.

pf: Consider the cover $\{B_n(p) \mid n \in \mathbb{N}\}$ for some $p \in K$. Since K is cmpt, $K \subset B_{n_1}(p) \cup \dots \cup B_{n_r}(p)$. Let $M = \max\{n_1, \dots, n_r\}$. Then $K \subset B_M(p)$. Given $q_1, q_2 \in K$, the triangle inequality gives $d(q_1, q_2) \leq d(q_1, p) + d(q_2, p) = 2M$.

p $13\frac{1}{2}$ First

Heine-Borel (1st version): A set $K \subset \mathbb{R}$ is cmpt if and only if it is closed and bounded.

Pf: $\Rightarrow$ already done.

$\Leftarrow$ Start by showing that any $[a,b] \subset \mathbb{R}$ is cmpt.

Thm: Let $\mathcal{U} \subset \mathbb{R}$ be open. Then $\mathcal{U}$ is the union of at most countably many disjoint open intervals.

Proof: Pick $x \in \mathcal{U}$ and let

$$a_x = \inf\{y, (y,x) \subset \mathcal{U}\}$$
$$b_x = \sup\{y, (x,y) \subset \mathcal{U}\}$$

Then $(a_x, b_x) \subset \mathcal{U}$. If not, then $a_x \; (\; x \;) \; b_x$

Next let $\hat{x} \in (a_x, b_x)$. Then $(a_{\hat{x}}, b_{\hat{x}}) = (a_x, b_x)$. (If, say, $b_{\hat{x}} > b_x$, then $(a_{\hat{x}}, b_{\hat{x}}) \cap (a_x, b_x) \neq \phi$ $(\ni \hat{x})$, so $(a_{\hat{x}}, b_{\hat{x}}) \cup (a_x, b_x) \supset (a_x, b_{\hat{x}})$. Hence b_x could not be the sup.) Thus any two intervals thus constructed must be disjoint and their union must be $\mathcal{U}$.

Finally, each interval contains a rational number. There are countably many of these $\Rightarrow$ countably many intervals.

Let $\{U_\beta\}$, $U_\beta \subset \mathbb{R}$ be an open cover of $[a,b]$. Now each U_β is a union of open intervals, so can assume that $[a,b]$ is covered by $\{O_\alpha = (a_\alpha, b_\alpha)\}$, where each $(a_\alpha, b_\alpha) \subset U_\beta$ for some β.

Now continue as on p. 15 to show that $[a,b]$ is covered by finitely many $O_{\alpha_1}, O_{\alpha_2}, \ldots, O_{\alpha_n}$ of the intervals.

Finally, choose U_{β_i} s.t. $O_{\alpha_i} \subset U_{\beta_i}$, $i = 1, 2, \ldots, n$. Then $[a,b] \subset U_{\beta_1} \cup U_{\beta_2} \cup \cdots \cup U_{\beta_n}$. Thus $[a,b]$ is cmpt.

see p. 15.1!

Let $\{O_\alpha = (a_\alpha, b_\alpha)\}$ be a collection of open intervals covering $[a,b]$.

Let $S = \{c \mid a \le c \le b,\ [a,c]$ is covered by finitely many of the $O_\alpha\}$

Obviously $S \neq \emptyset$. S is also bounded, so $d = \sup S$ is well defined.

Now assume $d < b$. Then $[a,d]$ must be covered by finitely many of the O_α. Let O_β be such that $d \in O_\beta$. Then there are $c > d$ s/t $[a,c]$ is covered with finitely many of the O_α, which is a contradiction. Thus $d = b$ and $[a,b]$ is covered by finitely many of the O_α.

Next, let $E \subset \mathbb{R}$ be closed and bounded. Since E is bounded, $E \subset [a,b]$ for some a, b. Thus E is a closed subset of a cmpt set, so E is cmpt.

Let $f: X \to \mathbb{R}$, where f is cont. and X is compact. Then f is bounded and it attains its max and min values.

Pf: By the above, $f(X) \subset \mathbb{R}$ is cmpt so it must be bounded. Thus f is bounded.

Let $a = \inf\{f(x) \mid x \in X\}$ and $b = \sup\{f(x) \mid x \in X\}$. Since $f(X)$ is closed, $a, b \in f(X)$, that is, $a = f(p)$, $b = f(q)$ for some $p, q \in X$.

Thm: Let $f: X \to Y$, where X is cmpt and Y is Hausdorff. Suppose f is 1-to-1 and onto. Then f^{-1} is continuous, that is, f is a homeomorphism.

Proof: Let $C \subset X$ be closed. Then C is cmpt. Hence $f(C)$ is also cmpt and so it is closed. Thus f takes closed sets to closed sets, so f^{-1} must be continuous.

Bolzano - Weierstrass: An infinite subset of X, X cmpt, must have a limit point.

pf: Let $S \subset X$ have no limit points, S cmpt. Then given $x \in X$ there is an open nbhd O_x of x s/t

$$O_x \cap S = \begin{cases} \emptyset, & \text{if } x \notin S \\ \{x\}, & \text{if } x \in S \end{cases}$$

Now $\{O_x\}$ is an open cover of X, so ...
$X = O_{x_1} \cup \cdots \cup O_{x_n}$ for some O_{x_i}. Each O_{x_i} contains at most 1 point from S, so S must be finite.

Lebesgue Lemma: Let (X, d) be a cmpt metric space and let $\mathcal{F} = \{O_\alpha\}_{\alpha \in \mathcal{A}}$ be an open cover of X. Then there exist $\delta > 0$ such that any subset of diameter less than δ is entirely contained in some member of $\mathcal{F}$.

Diameter $\operatorname{diam}(A) = \sup\{d(p,q) \mid p, q \in A\}$.
δ in the above lemma is called a Lebesgue number of $\mathcal{F}$.

(*) pf: Assume on the contrary that there is a sequence $A_n \subset X$ of sets such that $\lim_n \operatorname{diam}(A_n) = 0$, and none of the A_n is contained in a single member of $\mathcal{F}$.

For each n, pick $x_n \in A_n$. Then the sequence

(*) See p. 18

Constructive proof of Lebesgue Lemma:

Let $\mathcal{F} = \{O_\alpha\}_{\alpha \in \mathcal{A}}$ be an open cover of X. For every $x \in X$, choose $O_{\alpha_x} \in \mathcal{F}$ s/t $x \in O_{\alpha_x}$. Since O_{α_x} is open, there is $r_x > 0$ s/t $B_{r_x}(x) \subset O_{\alpha_x}$.

Now the collection of balls $\{B_{r_x/2}(x), x \in X\}$ is a cover of X. By cmptness, there is a finite subcover $\{B_{r_{x_i}/2}(x_i), i = 1, 2, \ldots, n\}$ of X.

Let $\delta = \min\{r_{x_i}/2\}_{i=1,2,\ldots,n}$. Clearly $\delta > 0$.

Let $A \subset X$ with $\operatorname{diam}(A) < \delta$. Pick some $p \in A$. Then for some x_i, $p \in B_{r_{x_i}/2}(x_i)$. Let $q \in A$ be any. Then $d(p,q) \le \operatorname{diam}(A) < \delta \le r_{x_i}/2$.

Thus $d(x_i, q) \le d(x_i, p) + d(p, q) < r_{x_i}$, so $A \subset B_{r_{x_i}}(x_i) \subset O_{\alpha_{x_i}}$, which proves the lemma.

Uniform continuity:

Thm: Let $f: X \to Y$, where (X, d_X), (Y, d_Y) are metric spaces and X is compact. Given $\varepsilon > 0$ there is $\delta > 0$ s/t $f(B(x, \delta)) \subset B(f(x), \varepsilon)$ for all $x \in X$.

pf: Cover Y with $\{B(y, \varepsilon/2)\}_{y \in Y}$. Then $\{f^{-1}(B(y, \varepsilon/2))\}$ is an open cover of X. Let $\tilde{\delta} > 0$ be the Lebesgue number of this cover. Now each $B(x, \tilde{\delta}/2)$ is contained in one of $f^{-1}(B(y, \varepsilon/2))$, we have $f(B(x, \tilde{\delta}/2)) \subset B(y, \varepsilon/2)$

Let $\hat{x} \in B(x, \tilde{\delta}/2)$. Then $d_Y(f(\hat{x}), f(x))$
$\leq d_Y(f(\hat{x}), y) + d_Y(y, f(x)) < \varepsilon$.

Let $\delta = \tilde{\delta}/2$.

Proved: Given $\varepsilon > 0$, there is $\delta > 0$ s/t ...

Prop: X is compact $\iff$

Suppose $\mathcal{C} = \{C_\alpha\}_{\alpha \in A}$ is a family of closed sets with $\bigcap_{\alpha \in A} C_\alpha = \emptyset$.

Then there are $C_{\alpha_1}, \dots, C_{\alpha_n} \in \mathcal{C}$ s/t $\bigcap_{i=1}^{n} C_{\alpha_i} = \emptyset$.

Proof: $\Rightarrow$ If $\bigcap_{\alpha \in A} C_\alpha = \emptyset$, then

$\neg \bigcap_{\alpha \in A} C_\alpha = \bigcup_{\alpha \in A} \neg C_\alpha = X$. X compact,

so $X = \bigcup_{i=1}^{n} \neg C_{\alpha_i} \iff \bigcap_{i=1}^{n} C_{\alpha_i} = \emptyset$.

$\Leftarrow$ Let $\{O_\alpha\}_{\alpha \in A}$ be an open cover of X.

Then $\neg \bigcup_{\alpha \in A} O_\alpha = \bigcap_{\alpha \in A} \neg O_\alpha = \emptyset$

By assumption, $\bigcap_{i=1}^{n} \neg O_{\alpha_i} = \emptyset$ for some $\alpha_i \in A$.

Then $X = \bigcup_{i=1}^{n} O_{\alpha_i}$.

§ 3.5. Connectedness

Say X is disconnected if $X = A \cup B$, where $A, B \subset X$ are nonempty, open and disjoint.

If X is not disconnected, then call X connected.

Ex: $\mathbb{Q}$ is disconnected: $A = \{x \in \mathbb{Q} \mid x < \sqrt{2}\}$, $B = \{x \in \mathbb{Q} \mid x > \sqrt{2}\}$. Any $p, q \in \mathbb{Q}$ separable. (Show!)

Ex: X with discrete topology. (Totally disconnected.)

Prop: Let X be connected and let $f: X \to Y$ be cont. Then $f(X) \subset Y$ is connected.

Pf: Suppose not. Then $f(X) = A \cup B$, where ....
Now $X = f^{-1}(A) \cup f^{-1}(B)$ implies that X is disconn. $\lightning$

Cor: Let $f: X \to Y$ be a homeom. Then X is conn. if and only if Y is.

Note: X is connected iff $\emptyset, X$ are the only sets that are both open and closed.

Recall: Let $A \subset Y \subset X$ with Y equipped with the ssp topology. Then $\bar{A}^Y = \bar{A}^X \cap Y$. (*)

Suppose $Y = \mathring{A} \cup \mathring{B}$ (†), $A \cap B = \emptyset$. Then $A \subset Y$ is both open and closed. Now $\bar{A}^Y = \bar{A}^X \cap Y$, so $A = \bar{A}^Y = \bar{A}^X \cap Y \Rightarrow \bar{A}^X \cap B = \emptyset$.
Sim. $\bar{B}^X \cap A = \emptyset$.

Conversely, suppose $Y = A \cup B$, where $\bar{A}^X \cap B = A \cap \bar{B}^X = \emptyset$. Then $\bar{A}^X \cap Y = A$ and $\bar{B}^X \cap Y = B$. Hence both A & $B \subset Y$ are closed. Consequently, they are also open.

Conclusion: $Y \subset X$ is disconnected iff $Y = A \cup B$, where $\bar{A}^X \cap B = \emptyset$ and $A \cap \bar{B}^X = \emptyset$ (and $A, B \neq \emptyset$).

(†) $A, B \subset Y$ disjoint, non-empty, and open in the subspace topology.

(*) $A \subset \bar{A}^X \cap Y$, which is closed in $Y \Rightarrow \bar{A}^Y \subset \bar{A}^X \cap Y$.
Since $\bar{A}^Y \subset Y$ is closed, $\bar{A}^Y = Y \cap C$, $C \subset X$ closed.
Then $\bar{A}^X \subset C$, so $\bar{A}^X \cap Y \subset C \cap Y = \bar{A}^Y$.

Thm: Suppose $Z \subset X$ is connected, and let $Y \subset X$ satisfy $Z \subset Y \subset \overline{Z}$. Then Y is connected. In particular, if $Z \subset X$ is connected, then so is $\overline{Z}$.

Pf: Suppose, on the contrary, that Y is disconnected so that $Y = A \cup B$, where $A, B \subset Y$ are open, disjoint and non-empty. Write $A = Y \cap \hat{A}$, $B = Y \cap \hat{B}$, where $\hat{A}, \hat{B} \subset X$ are open.

Since $Z \subset Y$, $Z = (Z \cap \hat{A}) \cup (Z \cap \hat{B})$, where $Z \cap \hat{A}$, $Z \cap \hat{B}$ are open in Z and disjoint. But Z is connected, so either $Z \cap \hat{A}$ or $Z \cap \hat{B}$ must be empty. Suppose $Z \cap \hat{A} = \phi$. Since $\hat{A} \subset X$ is open, no point in $\hat{A}$ can be an accumulation point of Z. In particular, $\hat{A} \cap \overline{Z} = \phi$. But $Y \subset \overline{Z}$, so $A = Y \cap \hat{A} = \phi$, which contradicts our initial assumption.

The following is needed in the 2nd part of the Thm on p. 24-25

Prop: Let $A \subset \mathbb{R}$ s/t if $a, b \in A$, then $[a, b] \subset A$. Then A is an interval.

Pf: Let $a_0 = \inf A$, $b_0 = \sup A$, suppose $a_0, b_0 \notin A$. Let $a_0 < x < b_0$. Now there is $\{a_n\} \subset A$, $a_n \to a_0$. $\{b_n\} \subset A$, $b_n \to b_0$. Then some $a_N < x < b_N \Rightarrow x \in A$. So $A = (a_0, b_0)$. Other cases $a_0 \in A$ and/or $b_0 \in A$ sim.

Cor. A is not an interval if $\exists\, a, b \in A$ and $a < x < b$ s/t $x \notin A$.

Intervals: Definition.

$(a,b) = \{x \mid a<x<b,\ -\infty \le a,\ b \le \infty\}$

$[a,b) = \ldots$

$(a,b] = \ldots$

$[a,b] = \ldots$

Thm: A subset $A \subset \mathbb{R}$, $A \neq \phi$, is connected if and only if it is an interval.

Pf: To prove that any interval is connected, it suffices, by the above to show that any open interval is connected. (*) [Moreover, any two open intervals are homeomorphic, so wlog we can assume that $A = I = (0,1)$.] Now assume that $I = P \cup Q$, where .... Pick $p \in P$, $q \in Q$. Can assume $p<q$. Let $z = \sup\{[p,x] \subset P\}$, so $z \le q$. If $z \in Q$, then $(z-\varepsilon, z) \subset Q$ for some $\varepsilon > 0$ }. If $z \in P$, then $[z, z+\varepsilon) \subset P$ for some $\varepsilon > 0$ } cont'd p. 23

(*) Not needed.

Next assume that A is not an interval. Then there are $x, y \in A$, $z \in \neg A$ s/t $x < z < y$. Let $\mathcal{O} = A \cap (-\infty, z)$ and $\mathcal{P} = A \cap (z, \infty) \Rightarrow A$ is not connected. Ex: Intermediate value thm!

(*) Prop: Let $x \in X$ and let $\mathcal{U} = \{A \subset X,$ A is connected and $x \in A\}$. Let $B = \bigcup_{A \in \mathcal{U}} A$. Then B is connected.

Pf. Suppose not. Then $B = \mathcal{O} \cup \mathcal{P}$, where.... Can assume $x \in \mathcal{O}$, $x \notin \mathcal{P}$. Then given $A \in \mathcal{U}$, $A = (\mathcal{O} \cap A) \cup (\mathcal{P} \cap A)$, where $\mathcal{O} \cap A \neq \emptyset$. But A is connected, so $\mathcal{P} \cap A = \emptyset$ for all $A \in \mathcal{U}$. It follows that $\mathcal{P} = \emptyset$, so B is connected.

Cor: Any $x \in X$ is contained in a maximal connected subset of X.

Call these connected components. They are disjoint and closed. (Cf. $\mathbb{Q} \subset \mathbb{R}$)

(B connected $\Rightarrow \overline{B}$ connected. But B is maximal, so $B = \overline{B}$. Thus B is closed.)
If X has a finite number of connected comp., then these are also open.
Ex: $\mathbb{Q}$: connected comp?

(*) Cor: Suppose $A_i \subset X$, $i = 1, 2, \ldots$, are connected and $A_{i+1} \cap A_1 \cup A_2 \cup \cdots \cup A_i \neq \emptyset$ for all i. Then $\bigcup_{i=1}^{\infty} A_i$ is connected.